Showing that $\inf_{n\in\mathbb{Z}} \left\{\dfrac{n^2+1}{2n-1}\right\} = -\infty$

Intuition

Consider the function

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \frac{x^2+1}{2x-1}.$$

For large positive x, dividing numerator and denominator by x gives

$$f(x) = \frac{x + \frac{1}{x}}{2 - \frac{1}{x}} \approx \frac{x}{2},$$

so $f(x)$, when $|x|$ is large, behaves like the line $y = \frac{1}{2}x$. As $x \to -\infty$, both x and $2x-1$ are negative, so the same asymptotic holds:

$$f(x) \approx \frac{1}{2}x.$$

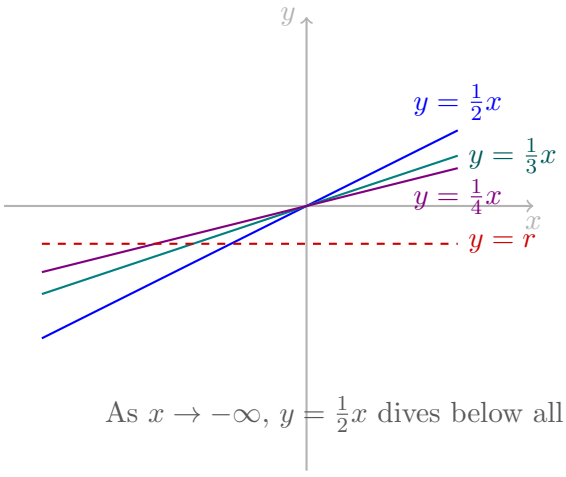
The line $y = \frac{1}{2}x$ has a steeper (more negative) slope than $y = \frac{1}{3}x$ or $y = \frac{1}{4}x$:

$$\tfrac{1}{2}x < \tfrac{1}{3}x < \tfrac{1}{4}x \quad \text{for large negative } x.$$

Hence, for large negative x, the graph of $y = \frac{1}{2}x$ lies *below* those of $y = \frac{1}{3}x$ and $y = \frac{1}{4}x$, and therefore $f(x)$ will eventually fall below any of them. This geometric picture explains why, for any real r, we can find $n_1 \in \mathbb{Z}_-$ such that

$$f(n_1) < \tfrac{1}{3}n_1 < r,$$

so $\inf f(n) = -\infty$. (I we can use any of $1/3x$ or $1/4x$ or a lot of different lines for an easy comparison.)



As $x \to -\infty$, $y = \frac{1}{2}x$ dives below all others.

Formal Argument

For $n \leq 0$, the denominator $2n - 1 < 0$. Compute:

$$f(n) - \frac{n}{3} = \frac{(n^2+1) - \frac{n}{3}(2n-1)}{2n-1} = \frac{\frac{1}{3}(n^2+n+3)}{2n-1}.$$

The numerator $\frac{1}{3}(n^2+n+3) > 0$ for when $|n|$ is very large (positive or negative), while the denominator $2n - 1 < 0$ for $n \leq 0$. Thus

$$\boxed{f(n) \leq \frac{n}{3} \quad \text{for all } n \leq 0.}$$

Now let $r \in \mathbb{R}$. By the Archimedean property, there exists a negative integer n_1 such that $\frac{n_1}{3} < r$. Then $f(n_1) \leq \frac{n_1}{3} < r$, proving that $f(n)$ takes values smaller than any real number r.

$$\boxed{\inf_{n \in \mathbb{Z}} f(n) = -\infty.}$$

Question. Can you use a similar argument to show that $\sup_{n \in \mathbb{Z}} \left\{ \frac{n^2+1}{2n-1} \right\} = +\infty$?